Petromyzon marinus (Sea Lamprey)
Fishes
Native Transplant

Collection Info
Point Map
Species Profile
Animated Map



Great Lakes Fishery Commission

Petromyzon marinus Linnaeus, 1758

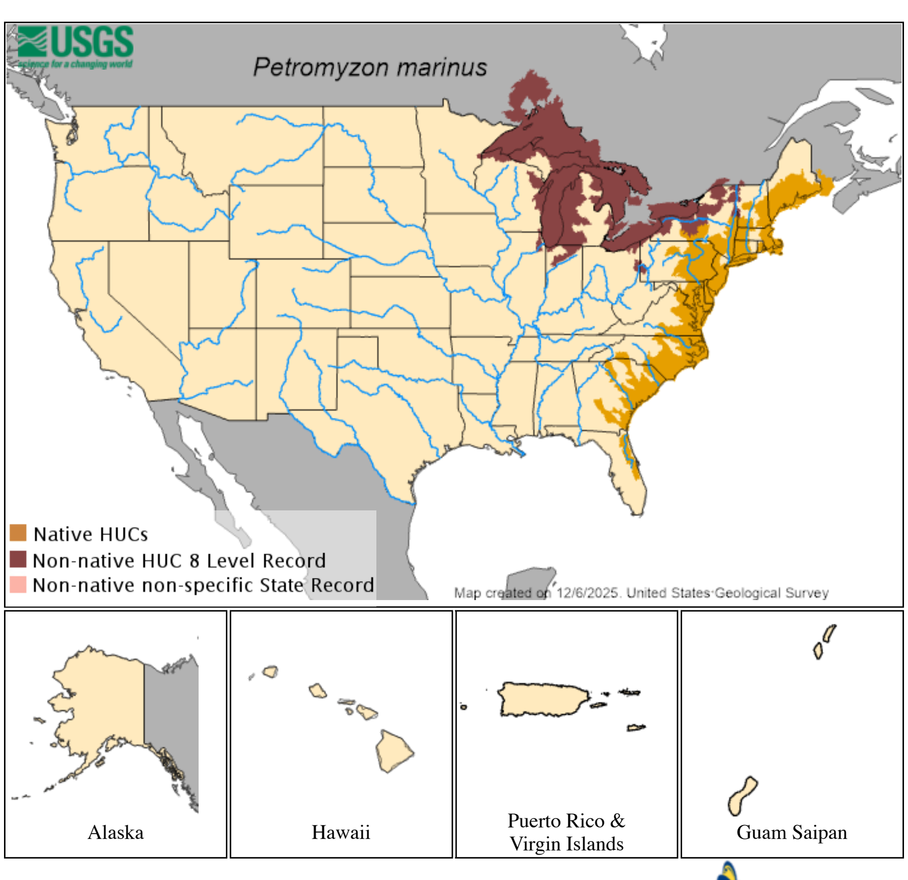
Common name: Sea Lamprey

Taxonomy: available through ITIS

Identification: The sea lamprey is a jawless cartilaginous fish that is somewhat eel-like in appearance. This species has two closely-spaced but separate dorsal fins, no paired fins, seven gill openings on each side of its head, and a large round sucker-like mouth ringed with small, sharp teeth that act as a rasp along with a file-like tongue. Juvenile parasitic sea lamprey are 6 to 24 inches in length with smooth, scaleless skin that is mottled gray-blue to black, darker on top and fading to a lighter colored belly. Adult sea lamprey are 14 to 24 inches in length and exhibit mottled dark brown/black pigmentation. Several keys to the ammocetes of lampreys found in the Great Lakes region are available from Becker (1983); Page and Burr (1991); Jenkins and Burkhead (1994); and Vladykov and Kott (1980). For further identification resources, see Page and Burr 1991; Jenkins and Burkhead 1994.

Size: 120 cm anadromous; 64 cm landlocked

Native Range: Generally marine but ascends freshwater rivers to spawn. Atlantic Coast from the Gulf of St. Lawrence to the St. Johns River, Florida; Atlantic Coast of Europe and Mediterranean Sea (Page and Burr 2011).



Native range data for this species provided in part by NatureServe
Hydrologic Unit Codes (HUCs) Explained
Interactive maps: Point Distribution Maps

Nonindigenous Occurrences:

Table 1. States with nonindigenous occurrences, the earliest and latest observations in each state, and the tally and names of HUCs with observations†. Names and dates are hyperlinked to their relevant specimen records. The list of references for all nonindigenous occurrences of *Petromyzon marinus* are found here.

State	First Observed	Last Observed	Total HUCs with observations†	HUCs with observations†
IL	1936	1998	2	Des Plaines; Lake Michigan
IN	1949	2014	4	Lake Michigan; Little Calumet-Galien; St. Joseph; Tippecanoe
MI	1934	2022	34	Au Gres-Rifle; Bad-Montreal; Betsie-Platte; Betsy-Chocolay; Black-Macatawa; Black-Presque Isle; Boardman-Charlevoix; Carp-Pine; Dead-Kelsey; Fishdam-Sturgeon; Kalamazoo; Keweenaw Peninsula; Lake Huron; Lake Michigan; Lake St. Clair; Lake Superior; Little Calumet-Galien; Lone Lake-Ocqueoc; Lower Grand; Manistee; Manistique River; Maple; Millecoquins Lake-Brevoort River; Muskegon; Ontonagon; Pere Marquette-White; St. Clair; St. Joseph; St. Marys; Sturgeon; Tacoosh-Whitefish; Tahquamenon; Thunder Bay; Waiska
MN	1946	2001	2	Beartrap-Nemadji; Lake Superior
NY	1863	2020	19	Ausable River; Black; Buffalo-Eighteenmile; Cattaraugus; Grass; Irondequoit-Ninemile; Lake Champlain; Lake Erie; Lake Ontario; Mettawee River; Niagara River; Oak Orchard-Twelvemile; Oneida; Raisin River-St. Lawrence River; Raquette; Salmon; Salmon-Sandy; Seneca; St. Regis
OH	1927	2020	4	Ashtabula-Chagrin; Chautauqua-Conneaut; Grand; Lake Erie
PA	1985	2013	3	Chautauqua-Conneaut; Lake Erie; Upper Ohio
WI	1958	2020	9	Bad-Montreal; Beartrap-Nemadji; Door-Kewaunee; Lake Michigan; Lake Superior; Lower Fox; Manitowoc-Sheboygan; Oconto; Peshtigo

Table last updated 12/8/2025

† Populations may not be currently present.

Ecology: Sea lampreys are an ancient species that have retained their primitive ancestral characteristics from millions of years ago. This species is diadromous, spending the early stages of their life in streams and rivers and the middle stage of their life in the ocean or, in the case of the Great Lakes populations, in a large freshwater lake. They then return as breeding adults to spawn in the freshwater streams and rivers, and die shortly after spawning. The blind, worm-like larval lamprey, known as ammocoetes, can grow up to 5 inches long. They hatch from eggs in gravel nests in tributaries and drift downstream with the current. When they locate suitable habitat -- usually silt or sand stream bottoms and banks in slower-moving stretches of water -- they burrow in and take up residence, filter-feeding on algae, detritus and microscopic organisms and materials (Lake Champlain Sea Lamprey Control, 2020). Lampreys may stay in this larval form from 3-17 years. Nutrition levels in larval habitat may influence sex assignment in this fish, with larvae that consume more nutrients being more likely to develop as female (Bircenau 2017).

As they mature, the larval lampreys grow eyes and a sucker-like mouth. Once the ammocoetes' transformation is complete, the sea lamprey leaves its burrow and moves downstream to open water. The sea lamprey is then ready to begin the next stage in its life as a parasite of fish: the juvenile sea lampreys move into deeper water and begin to seek host fish on which to feed over the course of 12-20 months. In the spring, adult sea lamprey migrate up tributaries to sexually mature and spawn. Spawning streams are located by following pheromones released by ammocoetes living in those waters. Once they reach a suitable spawning site, usually rocky riffle areas that are shallow with fairly swift current, male and female sea lampreys build a nest, often called a redd. The female lays tens of thousands of eggs and the male fertilizes them, after which both adults die. Weeks later, the eggs hatch and the life cycle of the sea lamprey begins again (Lake Champlain Sea Lamprey Control, 2020).

Means of Introduction: Controversy exists as to whether the sea lamprey is native to Lake Ontario. Several believe it is native (e.g., Lawrie 1970; Smith 1985), suggesting that sea lamprey found in Lake Ontario and its tributaries, the Finger Lakes, and Lake Champlain represent relict populations from the last Pleistocene glaciation. Those contending that it is not native believe that this species, unknown in Lake Ontario prior to the 1830s, had most likely entered the inland lake from Atlantic coastal drainages via the artificially created Erie Canal (e.g., Emery 1985). Whether or not the sea lamprey is native to Lake Ontario, this species is not native to the other Great Lakes and tributaries where it is now readily found. The sea lamprey was previously prevented from spreading into Lake Erie and the rest of the Great Lakes basin by Niagara Falls. The Welland Canal, opened in 1829, bypassed Niagara Falls providing a route to Lake Erie from Lake Ontario (Aron and Smith 1971). From the opening of the Welland Canal (1829) to the discovery of sea lamprey in Lake Erie (1921), there is almost a century difference. Yet sea lamprey was found throughout the Great Lakes to the farthest Great Lake, Lake Superior, within twenty-five years of their arrival to Lake Erie. The improvements done to the Welland Canal in 1919 are likely the change that facilitated sea lamprey immigration into Lake Erie (1921) (Great Lakes Fishery Commission, 2019).

Status: Widespread populations overwinter and reproduce in tributaries throughout the Great Lakes basin. This species was common in Lakes Michigan and Huron by the 1930s and in eastern Lake Superior by the 1940s; abundance was initially low in Lake Ontario (Applegate 1950; Emery 1985) and Lake Erie (Smith 1985), but has fluctuated ever since. As an example: abundance in Lake Erie was historically low, increased substantially after 2004, and is currently dropping -- estimated to be around 20,000 individuals (Status of Sea Lamprey, 2019).

Impact of Introduction: Attack and parasitic feeding on other fishes by adult lampreys often results in death of the prey, either directly from the loss of fluids and tissues or indirectly from secondary infection of the wound (Phillips et al. 1982). Of the fish that survived attacks by lampreys, 85% of various species had been attacked up to five times (Scott and Crossman 1973). The species' introduction to the Great Lakes and its later abundance, combined with water pollution and overfishing, resulted in the decline of several large native species, including several ciscoes *Coregonus* spp., lake trout *Salvelinus namaycush*, and walleye *Sander vitreus*, among others. Consequently, there was a collapse in the commercial fisheries during the 1940s and 1950s in many parts of the Great Lakes, particularly in lakes Huron and Michigan, and in eastern Lake Superior (e.g., Lawrie 1970; Scott and Crossman 1973; Christie 1974; Lee et al. 1980 *et seq*.; Smith and Tibbles 1980; Becker 1983; Emery 1985; Courtenay 1993). Lake trout catch in Lake Huron fell from 3.4 million pounds in 1937 to virtual failure in 1947. In Lake Michigan, U.S. catch fell from 5.5 million pounds in 1946 to 402 pounds in 1953. In Lake Superior, catch dropped from an average of 4.5 million pounds to 368 thousand pounds in 1961 (Scott and Crossman 1973). In freshwater, sea lampreys are also known to attack white sucker *Catostomus commersoni*, longnose sucker *Catostomus catostomus*, redhorse *Moxostoma* spp., yellow perch *Perca flavescens*, rainbow trout *Oncorhynchus mykiss*, burbot *Lota lota*, channel catfish *Ictalurus punctatus*, northern pike *Esox lucius*, and common carp *Cyprinus carpio* (Scott and Crossman 1973). Sea lamprey predation, in combination with other factors (i.e., overfishing and hybridization with more common cisco species), led to the extinction of three endemics in the Great Lakes; the longjaw cisco *Coregonus alpenae*, the deepwater cisco *C. johannae*, and the blackfin cisco *C. nigripinnis* (Miller et al. 1989). During the late 1940s, the alewife *Alosa pseudoharengus* invaded the Great Lakes from the Atlantic Ocean. Because the sea lamprey had greatly reduced the population of large predators, alewife populations exploded and were followed by tremendous die-offs, resulting in additional changes to fish species composition in the lakes (Smith and Tibbles 1980). Sea lampreys also took a toll on the introduced salmon in the Great Lakes, much to the dismay of anglers and state fish agencies. Although the number of sea lamprey in the Great Lakes has been reduced, they still kill substantial numbers of lake trout in some areas and thus are impeding the rebuilding of established populations (Schneider et al. 1996, and references therein).

Remarks: Early methods to control this species included mechanical weirs and electrical barriers (Scott and Crossman 1973; Smith and Tibbles 1980). Beginning in the late 1950s, sea lampreys began to be successfully controlled by use of the lampricide 3-trifluoromethyl-4-nitrophenol (TFM), a chemical agent that kills larval lampreys in their stream habitats (Smith and Tibbles 1980). The lampricide has reduced the population by over 90% of the 1961 peak (Scott and Crossman 1973). As a result, commercial fisheries reportedly have shown some recovery (Smith 1985; Page and Laird 1993) and the sea lamprey's impact on native fishes has been reduced (Page and Laird 1993). However, continued use of TFM is apparently required to keep sea lamprey populations under control (Scott and Crossman 1973; Becker 1983). TFM is sometimes harmful to other fish (e.g., walleye) (Becker 1983), as well as to the larvae of nonparasitic lamprey species.

Results of an international symposium on the sea lamprey were published in the Canadian Journal of Fisheries and Aquatic Sciences in 1980. The demise of lake trout led to development of the splake, a hybrid between lake trout and brook trout. It was hoped that the hybrid would better avoid lampreys and mature faster, hence spawn at least once before becoming parasitized (Scott and Crossman 1973).

As of 1991, it was estimated that the U.S. and Canada were spending $8 million per year on lamprey control and another $12 million per year on lake trout restoration (Newman 1991). The sea lamprey is one of the most important invasive species in the Great Lakes. Although perhaps the first invader to the Great Lakes, having migrated out of Lake Ontario in the 1830s and into the other Great Lakes through the Welland Canal, it was not until the 1950s that impacts on fisheries were so great as to prompt serious management efforts. It was then that the Great Lakes Fisheries Investigations, the progenitor of the present day USGS Great Lakes Science Center in Ann Arbor was charged with discovering a lampricide targeting the larval stage. The Great Lakes Fisheries Commission was also formed at this time, an organization primarily concerned with sea lamprey management. Successful application of lampricide ensued, reducing the lamprey population dramatically. However, decimation by sea lamprey of predatory fish representing the top trophic levels of the Great Lakes food web had already caused another invader, the alewife, to proliferate. The alewife population exploded, causing fish kills that washed up on shore. Thus the sea lamprey created the need to control alewife. Due also tothe decimation of native predatory fish populations by sea lamprey, a Great Lakes sport fishery was created with the stocking of Chinook salmon in the 1960s.

There is currently a debate as to whether the sea lamprey are indigenous to Lake Ontario. Randy L. Eshenroder summarized the argument for and against this belief (Eshenroder, 2009, 2014). Analysis of mitochondrial DNA by Waldman et al. (2004, 2006) seem to show that sea lamprey is native to Lake Ontario and Lake Champlain. This hypothesis is contrary to the commonly held belief by Smith (1971) that the sea lamprey entered the lake by hitchhiking under boats after the construction of the Erie Canal in the 1820's. Smith proposes that the sea lampreys entered the Hudson River, went through the Erie Canal, entered the Oneida Lake, then entered the finger lakes and Lake Ontario. Waldman disagrees with Smith and maintains that the reason the sea lamprey was not seen before the canal was built is that it was rare due to cold water and lack of suitable habitat. He suggests that the population boomed after the canal was built because the environmental degradation associated with human settlement created a more hospitable habitat for the sea lamprey. There is much debate about the many haplotype differences seen between the Atlantic and Lake Ontario populations. Waldman believes that the presence of unique haplotypes in the lake (haplotypes B amd P) could not have developed in the short time between the opening of the Erie Canal and present time supporting the argument for nativity. Others suggest that the unique haplotypes observed could have simply gone extinct in the Atlantic or that more sampling needs to be done to find them. It is also possible that the sea lamprey could have invaded the lake more than once.

Waldman et al. (2009) wrote a comment in response to the issues raised above by Eshenroder (2009). Waldman did further genetic analysis on the Atlantic populations of sea lampreys and still maintains that the unique alleles found in Lake Ontario, but absent in the Atlantic coast collections, would have taken between 15,000 and 31,000 year to develop. He also refutes the hitchhiking under boats hypothesis by citing the difficult path the migrating lampreys would have had to follow including shallow pools, stagnant water, and complicated canal locks. Waldman also raises the issue that the "hitchhiking" behavior is actually a feeding behavior which is not exhibited in migrating lampreys - such as those entering the Hudson River. The lampreys would also be migrating in response to pheromonal signals sent from conspecifics upstream. These pheromones would be lacking in the Hudson River unless there was a population of sea lamprey already in the area. Eshenroder (2014) argues that *P. marinus* first entered Lake Ontario during a watershed breach between the Susquehanna River (in which lamprey are native) and Lake Ontario in 1863. The debate is yet unresolved.

References (click for full references)

Adair, R.A., and R.J. Young. 2007. Integrated management of sea lampreys in the Great Lakes 2006. Annual Report to the Great Lakes Fishery Commission, Ann Arbor, MI.

Adams, J. V., & Jones, M. L. (2020). Evidence of host switching: Sea lampreys disproportionately attack Chinook salmon when lake trout abundance is low in Lake Ontario. Journal of Great Lakes Research. doi:https://doi.org/10.1016/j.jglr.2020.03.003

Applegate, V.C. 1950. Natural history of the sea lamprey (Petromyzon marinus) in Michigan. U.S. Department of the Interior, Fish and Wildlife Service Special Scientific Report: Fisheries 55. 237 pp.

Aron, W.I., and S.H. Smith. 1971. Ship canals and aquatic ecosystems: equilibrium has not been achieved since the Erie, Welland, and Suez canals were built. Science 174:13-20.

Becker, G.C. 1983. Fishes of Wisconsin. University of Wisconsin Press, Madison, WI.

Bergstedt, R.A., and M.B. Twohey. 2007. Research to support sterile-male-release and genetic alteration techniques for sea lamprey control. Journal of Great Lakes Research 33(2):48-69.

Birceanu, O. (2017). To be or not to be...male or female? Journal of Experimental Biology, 220(15), 2684-2684. doi:10.1242/jeb.147520

Bracken, F. S. A., Rooney, S. M., Kelly-Quinn, M., King, J. J., & Carlsson, J. (2019). Identifying spawning sites and other critical habitat in lotic systems using eDNA "snapshots": A case study using the sea lamprey Petromyzon marinus L. Ecology and Evolution, 9(1), 553-567. doi:10.1002/ece3.4777

Brant, C. 2019. Great Lakes sea lamprey: the 70 year war on a biological invader. University of Michigan Press.

Bravener, G., & Twohey, M. (2016). Evaluation of a Sterile-Male Release Technique: A Case Study of Invasive Sea Lamprey Control in a Tributary of the Laurentian Great Lakes. North American Journal of Fisheries Management, 36(5), 1125-1138. doi:http://dx.doi.org/10.1080/02755947.2016.1204389

Bryan, M.B, D. Zalinski, B. Filcek, S. Libants, W. Li, and K.T. Scribner. 2005. Patterns of invasion and colonization of the sea lamprey. Molecular Ecology 14:3757–3773.

Byford, G. J., Wagner, C. M., Hume, J. B., & Moser, M. L. (2016). Do Native Pacific Lamprey and Invasive Sea Lamprey Share an Alarm Cue? Implications for Use of a Natural Repellent to Guide Imperiled Pacific Lamprey into Fishways. North American Journal of Fisheries Management, 36(5), 1090-1096. doi:10.1080/02755947.2016.1198286

Christie, W.J. 1974. Changes in the fish species composition of the Great Lakes. Journal of the Fisheries Research Board of Canada 31:827-854.

Courtenay, W.R., Jr. 1993. Biological pollution through fish introductions. 35-61 in B. N. McKnight, ed. Biological pollution: the control and impact of invasive exotic species. Proceedings of a symposium, Indiana University-Purdue University, Indiana Academy of Science, Indianapolis, IN.

Cudmore-Vokey, B., and E.J. Crossman. 2000. Checklists of the fish fauna of the Laurentian Great Lakes and their connecting channels. Canadian Manuscript Report of Fisheries and Aquatic Sciences 2500: v + 39p.

Dennis, C. E., Wright, A. W., & Suski, C. D. (2016). Potential for carbon dioxide to act as a non-physical barrier for invasive sea lamprey movement. Journal of Great Lakes Research, 42(1), 150-155. doi:10.1016/j.jglr.2015.10.013

Dunlop, E. S., McLaughlin, R., Adams, J. V., Jones, M., Birceanu, O., Christie, M. R., . . . Wilkie, M. P. (2018). Rapid evolution meets invasive species control: the potential for pesticide resistance in sea lamprey. Canadian Journal of Fisheries and Aquatic Sciences = Journal Canadien des Sciences Halieutiques et Aquatiques, 75(1), 152-168. doi:http://dx.doi.org/10.1139/cjfas-2017-0015

Eddy, S., and J.C. Underhill. 1974. Northern fishes, with special reference to the Upper Mississippi Valley, 3rd edition. University of Minnesota Press, Minneapolis, MN.

Elliot, R.F., and B.J. Gunderman. 2008. Assessment of remnant lake sturgeon populations in the Green Bay Basin, 2002-2006. Final Report to the Great Lakes Fishery Trust. Project Number 2001.113/2004.610. (Available online: http://www.fws.gov/midwest/sturgeon/)

Emery, L. 1985. Review of fish introduced into the Great Lakes, 1819-1974. Great Lakes Fishery Commission Technical Report, volume 45.

Eshenroder, R.L. 2009. Comment: mitochondrial DNA analysis indicates sea lampreys are indigenous to Lake Ontario. Transactions of the American Fisheries Society 138:1178-1189.

Evans, T. M. (2017). Are lampreys homebodies? Studying ammocoetes with open population models. Ecology of Freshwater Fish, 26(2), 168-180. doi:10.1111/eff.12261
Fetterolf, C.M., Jr. 1980. Why a Great Lakes Fishery Commission and why a Sea Lamprey International Symposium. Canadian Journal of Fisheries and Aquatic Sciences 37: 1588-1593.

Fisheries and Oceans Canada (FOC). 2009. Sea Lamprey Control. http://www.dfo-mpo.gc.ca/regions/central/pub/bayfield/06-eng.htm. Accessed 26 July 2012.

Francis, G.R., J.J. Magnuson, H.A. Regier, and D.R. Talhelm. 1979. Rehabilitating Great Lakes ecosystems. Great Lakes Fishery Commission. Tech. report no. 37.

Gambicki, S., & Steinhart, G. B. (2017). Changes in sea lamprey size and fecundity through time in the Great Lakes. Journal of Great Lakes Research, 43(3), 209-214. doi:10.1016/j.jglr.2017.03.014

Gerking, S.D. 1955. Key to the fishes of Indiana. Investigations of Indiana Lakes and Streams 4:49-86.

Great Lakes Fish Health committee (GLFHC). 2000a. Sterile-male-release technique. An innovative sea lamprey control method. Fact sheet 6, Ann Arbor, MI.

Great Lakes Fish Health committee (GLFHC). 2000b. Sea lamprey barriers. New technologies help solve an old problem. Fact sheet 5, Ann Arbor, MI.

Great Lakes Fishery Commission. 2012. How are Sea Lampreys Controlled. Available http://www.glfc.org/sealamp/how.php#trap Accessed 2 August 2012.

Gunderson, J. - Minnesota Sea Grant, Duluth, MN.

Hubert, T. D., Miller, J., & Burkett, D. (2019). A Brief Introduction to Integrated Pest Management for Aquatic Systems. North American Journal of Fisheries Management. doi:10.1002/nafm.10331

Hume, J. B., Meckley, T. D., Johnson, N. S., Luhring, T. M., Siefkes, M. J., & Wagner, C. (2015). Application of a putative alarm cue hastens the arrival of invasive sea lamprey (Petromyzon marinus) at a trapping location. Canadian Journal of Fisheries and Aquatic Sciences/Journal Canadien des Sciences Halieutiques et Aquatiques, 72(12), 1799-1806. doi:http://dx.doi.org/10.1139/cjfas-2014-0535

Imre, I., Di Rocco, R., Brown, G., & Johnson, N. (2016). Habituation of adult sea lamprey repeatedly exposed to damage-released alarm and predator cues. Environmental Biology of Fishes, 99(8/9), 613-620. doi:10.1007/s10641-016-0503-z

Jelks, H.L., S.J. Walsh, N.M. Burkhead, S. Contreras-Balderas, E. Diaz-Pardo, D.A. Hendrickson, J. Lyons, N.E. Mandrak, F. McCormick, J.S. Nelson, S.P. Platania, B.A. Porter, C.B. Renaud, J.J. Schmitter-Soto, E.B. Taylor, and M.L. Warren. 2008. Conservation status of imperiled North American freshwater and diadromous fishes. Fisheries 33(8):372-407.

Jenkins, R.E., and N.M. Burkhead. 1994. Freshwater Fishes of Virginia. American Fisheries Society, Bethesda, MD.

Jensen, A. J., & Jones, M. L. (2018). Forecasting the response of Great Lakes sea lamprey (Petromyzon marinus) to barrier removals. Canadian Journal of Fisheries & Aquatic Sciences, 75(9), 1415-1426. doi:10.1139/cjfas-2017-0243

Johnson, N. S., Lewandoski, S. A., Alger, B. J., O'Connor, L., Bravener, G., Hrodey, P., . . . Siefkes, M. J. (2020). Behavioral Responses of Sea Lamprey to Varying Application Rates of a Synthesized Pheromone in Diverse Trapping Scenarios. Journal of Chemical Ecology, 46(3), 233-249. doi:10.1007/s10886-020-01151-z

Johnson, N. S., Miehls, S., O'Connor, L. M., Bravener, G., Barber, J., Thompson, H., . . . Bruning, T. (2016). A portable trap with electric lead catches up to 75% of an invasive fish species. Scientific Reports, 6. doi:10.1038/srep28430

Johnson, N. S., Swink, W. D., & Brenden, T. O. (2017). Field study suggests that sex determination in sea lamprey is directly influenced by larval growth rate. Proceedings of the Royal Society B: Biological Sciences, 284(1851), 1-5. doi:10.1098/rspb.2017.0262

Jones, M. 2007. Toward improved assessment of sea lamprey population dynamics in support of cost-effective sea lamprey management. Journal of Great Lakes Research 33(Special Issue 2): 35-47.

Lawrie, A.H. 1970. The sea lamprey in the Great Lakes. Transactions of the American Fisheries Society 99:766-775.

Lee, D.S., C.R. Gilbert, C.H. Hocutt, R.E. Jenkins, D.E. McAllister, and J.R. Stauffer, Jr. 1980 et seq. Atlas of North American Freshwater Fishes. North Carolina State Museum of Natural History, Raleigh, NC.

Lennox, R. J., Bravener, G. A., Hsien-Yung, L., Madenjian, C. P., Muir, A. M., Remucal, C. K., . . . Cooke, S. J. (2020). Potential changes to the biology and challenges to the management of invasive sea lamprey Petromyzon marinus in the Laurentian Great Lakes due to climate change. Global Change Biology, 26(3), 1118-1137. doi:http://dx.doi.org/10.1111/gcb.14957

MacEachen, D.C., R.W. Russell, D.M. Whittle. 2000. Spatial distribution of mercury and organochloride contaminants in Great Lakes Sea Lamprey (Petromyzon marinus). Journal of Great Lakes Research 26(1): 112-119.

Madenjian, C.P., and T.J. Desorcie. 2010. Lake trout population dynamics in the northern refuge of Lake Michigan: implications for future rehabilitation. North American Journal of Fisheries Management 30(3):629-641.

Madenjian, C.P., B.D. Chipman, and J.E. Marsden. 2008. New estimates of lethality of sea lamprey (Petromyzon marinus) attacks on lake trout (Salvelinus namaycush): implications for fisheries management. Canadian Journal of Fisheries and Aquatic Sciences 65:535-642.

McCann, E. L., Johnson, N. S., Hrodey, P. J., & Pangle, K. L. (2018). Characterization of Sea Lamprey Stream Entry Using Dual-Frequency Identification Sonar. Transactions of the American Fisheries Society, 147(3), 514-524. doi:10.1002/tafs.10052

McLean, A. R., & McLaughlin, R. L. (2018). Consistent individual differences in sea lamprey (Petromyzon marinus) behaviour: Implications for control via trapping. Journal of Great Lakes Research, 44(3), 482-490. doi:10.1016/j.jglr.2018.03.002

McLeod, D.V., R.A. Cottrill, and Y.E. Morbey. 2011. Sea lamprey wounding in Canadian water of Lake Huron from 2000-2009: temporal changes differ among regions. Journal of Great Lakes Research 37: 601-608.

Miehls, S. M., Johnson, N. S., & Hrodey, P. J. (2017). Test of a Nonphysical Barrier Consisting of Light, Sound, and Bubble Screen to Block Upstream Movement of Sea Lampreys in an Experimental Raceway. North American Journal of Fisheries Management, 37(3), 660-666. doi:10.1080/02755947.2017.1308892

Miehls, S., Sullivan, P., Twohey, M., Barber, J., & McDonald, R. (2020). The future of barriers and trapping methods in the sea lamprey (Petromyzon marinus) control program in the Laurentian Great Lakes. Reviews in Fish Biology & Fisheries, 30(1), 1-24. doi:10.1007/s11160-019-09587-7

Miller, R.R., J.D. Williams, and J.E. Williams. 1989. Extinctions of North American fishes during the past century. Fisheries 14(6):22-38.

Milt, A. W., Diebel, M. W., Doran, P. J., Ferris, M. C., Herbert, M., Khoury, M. L., . . .

McIntyre, P. B. (2018). Minimizing opportunity costs to aquatic connectivity restoration while controlling an invasive species. Conservation Biology, 32(4), 894-904. doi:http://dx.doi.org/10.1111/cobi.13105

Minnesota Department of Natural Resources (MDNR). 2012. Minnesota invasive species laws. http://www.dnr.state.mn.us/eco/invasives/laws.html#prohibited. Accessed 28 July 2012.

Muhametsafina, A., Birceanu, O., Hlina, B. L., Tessier, L. R., & Wilkie, M. P. (2019). Warmer waters increase the larval sea lamprey's (Petromyzon marinus) tolerance to the lampricide 3-trifluoromethyl-4-nitrophenol (TFM). Journal of Great Lakes Research, 45(5), 921-933. doi:10.1016/j.jglr.2019.07.011

Myles-Gonzalez, E., McLean, A., & McLaughlin, R. (2019). Is there an impact of behaviour on trapping migratory invasive sea lamprey (Petromyzon marinus)? Aquatic Invasions, 14(3), 531-550. doi:10.3391/ai.2019.14.3.11

Nepszy, S.J. 1988. Parasites of fishes in Canadian waters of the Great Lakes. Technical Report No. 51. Great Lakes Fishery Commission, Ann Arbor, MI.
New York State Department of Environmental Conservation (NYSDEC). 2012. Lake Champlain Sea lamprey control. http://www.dec.ny.gov/animals/7240.html. Accessed 26 July 2012.

Newman, R.M. 1991. The invasion of foreign aquatic plants and animals. Minnesota-Out-Of-Doors 1991(May/June):16-17, 29.

Office of Technology Assessment (OTA), U.S. Congress. 1993. Harmful Non-Indigenous Species in the United States, OTA-F-565 (Washington, DC: U.S. Government Printing Office, September 1993).

Page, L.M., and B.M. Burr. 1991. A Field Guide to Freshwater Fishes of North America North of Mexico. The Peterson Field Guide Series, volume 42. Houghton Mifflin Company, Boston, MA.

Page, L.M., and B.M. Burr. 2011. Peterson Field Guide to Freshwater Fishes of North America North of Mexico. Second edition. Houghton Mifflin Harcourt Publing Company, New York, NY.

Page, L.M., and C.A. Laird. 1993. The identification of the nonnative fishes inhabiting Illinois waters. Report prepared by Center for Biodiversity, Illinois Natural History Survey, Champaign, for Illinois Department of Conservation, Springfield. Center for Biodiversity Technical Report 1993(4).

Patrick, H.K., T.M. Sutton, and W.D. Swink. 2009. Lethality of sea lamprey parasitism on lake sturgeon. Transactions of the American Fisheries Society 138:1065-1075.

Phillips, E.C., M.E.Washek, A.W. Hertel, and B.M. Niebel. 2003. The round goby (Neogobius melanostomus) in Pennsylvania tributary streams of Lake Erie. Journal of Great Lakes Research 29(1): 34-40.

Phillips, G.L., W.D. Schmid, and J.C. Underhill. 1982. Fishes of the Minnesota Region. University of Minnesota Press, Minneapolis, MN.

Rocco, R., Imre, I., Johnson, N., & Brown, G. (2016). Behavioural response of adult sea lamprey (Petromyzon marinus) to predator and conspecific alarm cues: evidence of additive effects. Hydrobiologia, 767(1), 279-287. doi:10.1007/s10750-015-2508-6

Rooney, S., xe, n, M., Wightman, G., xd, Conch, . . . King, J. J. (2015). Behaviour of sea lamprey (Petromyzon marinus L.) at man-made obstacles during upriver spawning migration: use of telemetry to assess efficacy of weir modifications for improved passage. Biology and Environment: Proceedings of the Royal Irish Academy, 115B(2), 125-136. doi:10.3318/bioe.2015.14

Schloesser, N. A., Merkes, C. M., Rees, C. B., Amberg, J. J., Steeves, T. B., & Docker, M. F. (2018). Correlating sea lamprey density with environmental DNA detections in the lab. Management of Biological Invasions, 9(4), 483-495. doi:10.3391/mbi.2018.9.4.11

Schneider, C.P., R.W. Owens, R.A. Bergstedt, and R. O'Gorman. 1996. Predation by sea lamprey (Petromyzon marinus) on lake trout (Salvelinus namaycush) in southern Lake Ontario, 1982-1992. Canadian Journal of Fisheries and Aquatic Sciences 53(9):1921-1932.

Scott, W.B., and E.J. Crossman. 1973. Freshwater Fishes of Canada. Fisheries Research Board of Canada, Bulletin 184. Ottawa.

Sherburne, S., & Reinhardt, U. G. (2016). First test of a species-selective adult sea lamprey migration barrier. Journal of Great Lakes Research, 42(4), 893-898. doi:10.1016/j.jglr.2016.04.009

Siefkes, M. J. (2017). Use of physiological knowledge to control the invasive sea lamprey (Petromyzon marinus) in the Laurentian Great Lakes. Conservation Physiology, 5. doi:10.1093/conphys/cox031

Siefkes, M.J. 2009. Sea lamprey. 50-57 in State of the Great Lakes 2009. U.S. Environmental Protection Agency and Environment Canada.

Smith, B.R. 1971. Sea lampreys in the Great Lakes of North America. 207-247 in Hardisty, M.W., and I.C. Potter, eds. The Biology of Lampreys. Vol. 1. Academic Press, New York.

Smith, B.R., and J.J. Tibbles. 1980. Sea lamprey (Petromyzon marinus) in Lakes Huron, Michigan, and Superior: history of invasion and control, 1936-78. Canadian Journal of Fisheries and Aquatic Sciences 37(11):1780-1801.

Smith, C.L. 1985. The Inland Fishes of New York State. New York State Department of Environmental Conservation, Albany, NY.

Smith, P.W. 1979. The Fishes of Illinois. University of Illinois Press, Urbana, IL.

Sotola, V. A., Miehls, S. M., Simard, L. G., & Marsden, J. E. (2018). Lateral and vertical distribution of downstream migrating juvenile sea lamprey. Journal of Great Lakes Research, 44(3), 491-496. doi:10.1016/j.jglr.2018.03.004

Swink, W.D. 2003. Host selection and lethality of attacks by sea lampreys (Petromyzon marinus) in laboratory studies. Journal of Great Lakes Research 29(Supp. 1):307-319. (Available online:http://www.epa.gov/solec/sogl2009/0018sealamprey.pdf)

Tessier, L. R., Long, T. A. F., & Wilkie, M. P. (2018). Influence of body size, metabolic rate and life history stage on the uptake and excretion of the lampricide 3-trifluoromethyl-4-nitrophenol (TFM) by invasive sea lampreys (Petromyzon marinus). Aquatic Toxicology, 194, 27-36. doi:10.1016/j.aquatox.2017.10.020

Thresher, R. E., Jones, M., & Drake, D. A. R. (2019). Evaluating active genetic options for the control of sea lamprey (Petromyzon marinus) in the Laurentian Great Lakes. Canadian Journal of Fisheries & Aquatic Sciences, 76(7), 1186-1202. doi:10.1139/cjfas-2018-0153

Tilmant, J.T. 1999. Management of nonindigenous aquatic fish in the U.S. National Park System. National Park Service.

Trautman, M.B. 1981. The Fishes of Ohio. Ohio State University Press, Columbus, OH.

Vladykov, V.D., and E. Kott. 1980. Description and key to metamorphosed specimens and ammocoetes of Petromyzonidae found in the Great Lakes region. Canadian Journal of Fisheries and Aquatic Sciences 37(11):1616-1625.

Wagner, C. M., Hanson, J. E., Meckley, T. D., Johnson, N. S., & Bals, J. D. (2018). A simple, cost-effective emitter for controlled release of fish pheromones: Development, testing, and application to management of the invasive sea lamprey. Plos One, 13(6). doi:10.1371/journal.pone.0197569

Waldman, J., R. Daniels, M. Hickerson, and I. Wirgin. 2009. Mitochondrial DNA analysis indicates sea lampreys are indigenous to Lake Ontario: response to comment. Transactions of the American Fisheries Society 138: 1190-1197.

Waldman, J.R., C. Grunwald, and I.I. Wirgin. 2006. Evaluation of the native status of sea lamprey Petromyzon marinus in Lake Champlain based on mitochondrial DNA sequencing analysis. Transactions of the American Fisheries Society 135:1076-1085.

Waldman, J.R., C. Grunwald, N.K. Roy, and I.I. Wirgin. 2004. Mitochondrial DNA analysis indicates sea lampreys are indigenous to Lake Ontario. Transactions of the American Fisheries Society 133:950-960.

Wilkie, M. P., Hubert, T. D., Boogaard, M. A., & Birceanu, O. (2019). Control of invasive sea lampreys using the piscicides TFM and niclosamide: Toxicology, successes & future prospects. Aquatic Toxicology, 211, 235-252. doi:10.1016/j.aquatox.2018.12.012

World Conservation Monitoring Centre. 1996. Coregonus alpenae. In: IUCN 2010. IUCN Red List of Threatened Species. Version 2010.4. Accessed 10 May 2011. (Available online: http://www.iucnredlist.org)

Xu, Y., Zhu, S. W., & Li, Q. W. (2016). Lamprey: a model for vertebrate evolutionary research. Zoological research, 37(5), 263–269. https://doi.org/10.13918/j.issn.2095-8137.2016.5.263

Zielinski, D. P., & Freiburger, C. (2020). Advances in fish passage in the Great Lakes basin. Journal of Great Lakes Research. doi:https://doi.org/10.1016/j.jglr.2020.03.008

Zielinski, D. P., McLaughlin, R., Castro-Santos, T., Paudel, B., Hrodey, P., & Muir, A. (2019). Alternative Sea Lamprey Barrier Technologies: History as a Control Tool. Reviews in Fisheries Science & Aquaculture, 27(4), 438-457. doi:10.1080/23308249.2019.1625300

Other Resources:
Sea Lamprey Fact Sheet (Great Lakes Fishery Commision)

Great Lakes Fishery Commission Homepage

Distribution in Illinois - ILNHS

Petromyzon marinus (Global Invasive Species Database)

Fishes of Wisconsin (Becker)

Great Lakes Waterlife

NY Invasive Species Clearinghouse

FishBase Summary

Author: Fuller, P., L. Nico, E. Maynard, J. Larson, A. Fusaro, and A.K. Bogdanoff

Revision Date: 12/18/2020

Peer Review Date: 8/25/2015

Citation Information:
Fuller, P., L. Nico, E. Maynard, J. Larson, A. Fusaro, and A.K. Bogdanoff, 2025, *Petromyzon marinus* Linnaeus, 1758: U.S. Geological Survey, Nonindigenous Aquatic Species Database, Gainesville, FL, https://nas.er.usgs.gov/queries/FactSheet.aspx?SpeciesID=836, Revision Date: 12/18/2020, Peer Review Date: 8/25/2015, Access Date: 12/8/2025

This information is preliminary or provisional and is subject to revision. It is being provided to meet the need for timely best science. The information has not received final approval by the U.S. Geological Survey (USGS) and is provided on the condition that neither the USGS nor the U.S. Government shall be held liable for any damages resulting from the authorized or unauthorized use of the information.